

Problem Statement & Objective

Adaptive 2D Systolic Array on PYNQ-Z2

Problem

Edge AI layers repeatedly multiply matrices and accumulate results.

Small or irregular shapes leave a fixed compute grid underused.

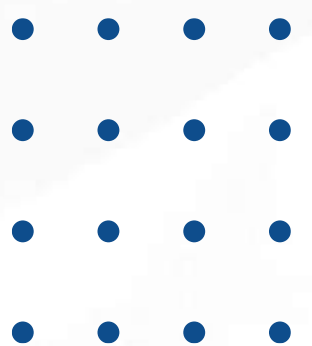
Data movement and FPGA routing constrain useful throughput.

Objective

Build exact signed INT8 acceleration with two parallel 16 x 16 engines.

Support active regions, zero padding and software tiling of larger matrices.

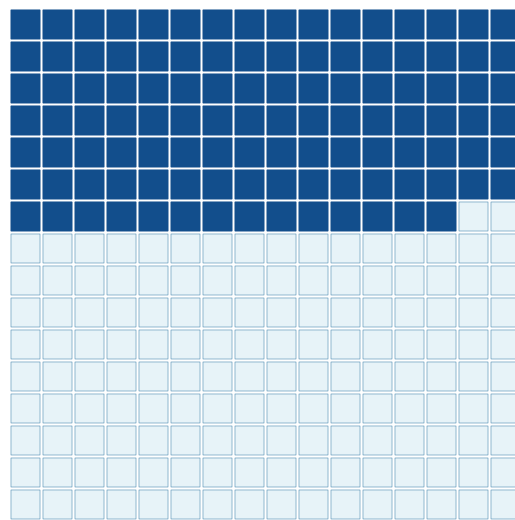
Verify correctness, timing and workload power before board release.



Proposed Solution

512 physical MAC lanes | Exact INT8 arithmetic | 100 MHz routed candidate

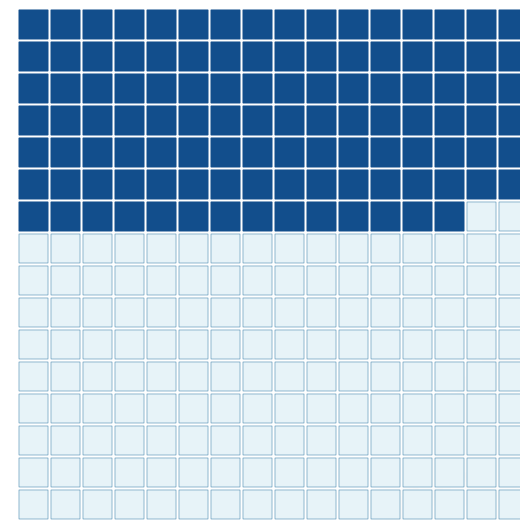
Engine 0



16 x 16 processing elements

110 DSP + 146 LUT MACs

Engine 1



16 x 16 processing elements

110 DSP + 146 LUT MACs

Arithmetic

Signed INT8 inputs; 20-bit tile accumulators; sign-extended INT32 output.

Adaptive execution

M / K / N up to 16 per tile. Active masks and zero padding handle tails; Python tiles larger matrices.

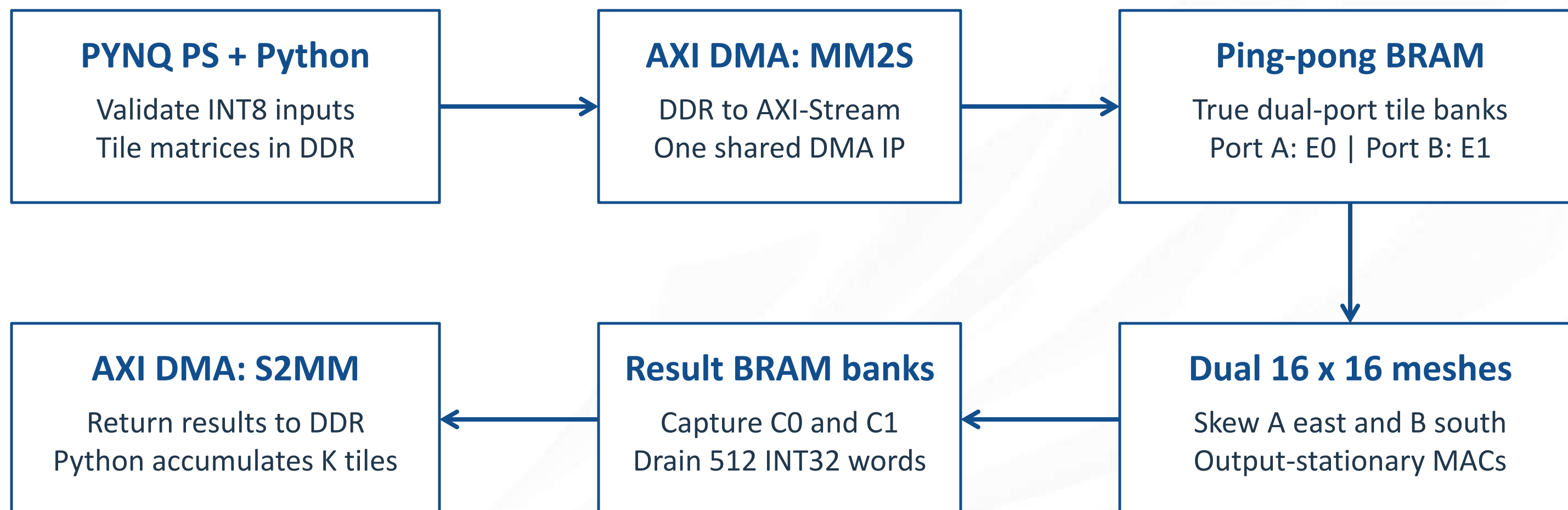
Shared data path

One AXI DMA; ping-pong true dual-port BRAM feeds both engines.

Compute ceiling: 51.2 GMAC/s at full utilization (calculated; throughput is unmeasured).

System Architecture & Workflow

Browser / gesture input -> PYNQ Python driver -> FPGA matrix computation



AXI4-Lite: dimensions, active masks, start/reset, status and cycle counters

HTTP bridge returns results and status to the browser; larger GEMMs use host tiling.

Innovation, Novelty & Value Addition

Workload adaptation

Runtime dimensions and active masks support small or irregular layers. Host tiling keeps the same hardware for larger GEMMs.

Useful on-chip storage

Two-port input banks map to BRAM. Row-enabled result reads and an elastic stream queue handle backpressure.

Balanced compute

110 DSP MACs in each engine plus 146 LUT MACs retain all 512 physical lanes with a symmetric allocation.

Measured implementation progress

Updated accelerator-only synthesis uses 31,899 LUTs versus 36,278 in the preserved baseline: 12.1% fewer LUTs.

Routed LUTs	Routed FFs	DSP48E1	BRAM36 equivalents
34,370 / 64.61%	33,133 / 31.14%	220 / 100%	26 / 18.57%

Full-board candidate: 12,043 / 13,300 slices (90.55%); the <75% slice objective remains unmet.

Software & Test Results

Software available

Python / PYNQ driver
DMA control, host tiling, INT32 accumulation
and cycle/time metrics.

Browser lab + HTTP bridge
Matrix editor, reference checks, dual-mesh
visualization and run status.

Gesture integration
Input and board-feedback scripts are present;
current board run is pending.

Verification	Recorded result
RTL regression	10 benches passed
Protocol regression	38 jobs; 20,239 checked words
Routed setup @ 100 MHz	WNS +0.188 ns; 0 failures
Routed hold / connectivity	WHS +0.010 ns; 0 unrouted nets
DRC	0 critical violations; warnings remain

Power and board acceptance are pending

0.897 W is a vectorless PL estimate. Workload activity, coverage review and physical-board validation are still required.

Team & Track Details

NEXT IN

Adaptive 2D Systolic Array Matrix Accelerator

Track 1 - FPGA | PYNQ-Z2 / Zynq-7000 XC7Z020

Processors - Problem #5: 2D Systolic Array-Based Processing Element

Team members: _____ Institution: _____

Thank You